

MKTG 565

Building Business Applications of LLMs and Generative Models

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Classroom	PCAR 295	Time: Wednesdays, 6:00–9:30 PM Spring 2026

Description

This course teaches you to *build* with large language models. You will start by experiencing what AI can do today through agentic systems such as coding agents. From there, you will develop the technical foundations to understand how it works and build your own LLM-powered prototypes from scratch. The main focus is demystifying generative AI so that future product managers, investors, and entrepreneurs can evaluate AI products, scope technical projects, and collaborate effectively with engineering teams.

Topics include attention mechanisms, embeddings, retrieval-augmented generation (RAG), fine-tuning, alignment, and agentic workflows, with applications in market research, content generation, AI search, and more. Through lectures, hands-on exercises, and a final project, you will build and evaluate working LLM applications and agentic systems. The emphasis is on feasibility, cost, risk, and on demoing what you build.

No prior coding experience needed. In the age of LLMs, English is the programming language—you describe what you want, and the AI writes the code. This course teaches you to build with that superpower.

This course is for anyone who wants to build with AI, not just talk about it—you will leave ready to prototype, evaluate, and lead AI-driven initiatives.

Learning Outcomes

By the end of this course, students will be able to:

1. Explain the core building blocks of generative AI (attention, embeddings, RAG, fine-tuning, alignment) and the AI product stack from chat interfaces to agentic workflows.
2. Build and prototype LLM-powered applications and agentic systems that address a real business problem.

3. Evaluate AI products and initiatives for feasibility, cost, risk, and responsible-AI considerations.

Course Materials

No textbook required. All materials are posted on Canvas.

- Lecture slides, Google Colab notebooks, case readings, articles, and class recordings are posted on Canvas.
- OpenAI API will be used in some course activities and assignments. \$5 should be sufficient.
- For vibe coding, you may use any AI coding tools that fit your workflow. No single tool is required for success in this course. IDE-based tools (e.g., Cursor, Antogravity) and CLI-based tools (e.g., Claude Code in the CLI or desktop app, Codex in the CLI or desktop app, Gemini CLI) are all fine.
- In class, I will primarily use Claude Code and Codex for live demos. If you already have a ChatGPT or Claude subscription, tools such as Codex or Claude Code may be a natural place to start. Some tools also offer free tiers, student access, or cloud credits, including [Cursor for Students](#), [Gemini CLI quota and pricing](#), and [GitHub Copilot for Students](#).

Course Schedule

Week	Date	Topic		
1	Apr. 1	Course Overview & Vibe Coding		
2	Apr. 8	LLM Vocabulary & API	<i>MiniAssignment 1</i>	<i>Major Assignment 1</i>
3	Apr. 15	Agentic AI		<i>MiniAssignment 2</i>
4	Apr. 22	Embeddings & RAG	<i>MiniAssignment 3</i>	<i>Major Assignment 2</i>
5	Apr. 29	GenAI Ecosystem & Midterm Check-in		
6	May 6	Fine-Tuning & Alignment		<i>Major Assignment 3</i>
7	May 13	GenAI Applications Part I		
8	May 20	GenAI Applications Part II		
		Guest speaker: Zheqing (Bill) Zhu Founder & CEO, Pokee AI Website LinkedIn		
9	May 27	AI Governance, Risks & Society		
		Guest speaker: Elan Ellinger Chief Revenue Operations, Global Business Applications, Microsoft LinkedIn		
10	Jun. 3	Demo Day		

Note: Assignment due dates are posted on Canvas. The assignments shown in the schedule indicate which week's content they are associated with, but their due dates may not fall on the same day as class.

Requirements

Class Participation

Every student should come prepared to contribute. Participation is assessed on quality, not quantity.

MiniAssignments

Three MiniAssignments (5% each, 15% total) are small individual assignments connected to the corresponding class week.

1. MiniAssignment 1: OpenAI API setup and first API call.
2. MiniAssignment 2: Agentic AI system showcase.
3. MiniAssignment 3: Embeddings and RAG.

Major Assignments

Three major assignments (10% each, 30% total) ask you to apply course concepts:

1. Major Assignment 1 (individual): LLM API & Batch Text Analysis. Build an automated pipeline using the OpenAI batch API to analyze product reviews at scale with zero-shot, few-shot, and chain-of-thought strategies.
2. Major Assignment 2 (group): RAG in the Wild. Choose your own data, build an end-to-end RAG pipeline, and evaluate whether RAG improves over a baseline.
3. Major Assignment 3 (group): Fine-Tuning. Fine-tune a model to generate structured outputs for a domain-specific business application.

Final Project

In the final project, your team will design, prototype, and present an end-to-end LLM-powered application that solves a business problem. The key expectation is a working prototype. Teams of 2 are encouraged, and individual projects are also allowed.

Project Checkpoints

1. Project Checkpoint 1 (Week 3). Submit your 1-page project plan on Canvas.
2. Project Checkpoint 2 (Week 5). In-class midterm check-in focused on your current direction, early progress, and any major questions or blockers. No separate submission is required.
3. Project Checkpoint 3 (Week 8). Submit a short progress update on Canvas, including your current prototype or demo status, with a link or screenshots if applicable.

Deliverables

1. Final Report (10 pages maximum). The report should clearly cover the business problem, the system you built, the data and implementation, the evaluation approach and evidence, and the main deployment trade-offs or limitations. Submit the complete final report after Demo Day.
2. Demo Day presentation. Present your project during the final session using a format that best communicates the working prototype.

Evaluation Criteria

These criteria are guiding dimensions; not every item applies equally to every project.

1. Problem Choice & Business Value (30%)
 - Is the business problem clearly defined and relevant?
 - Is the solution well-motivated, original enough to be interesting, and aligned with business goals?
 - Does the project demonstrate credible business value, practical relevance, and awareness of key deployment trade-offs or risks where relevant?
2. Technical Depth (35%)
 - Are the proposed techniques appropriate and well-justified?
 - Does the prototype function as intended?
3. Evaluation & Evidence (15%)
 - Are success metrics well-defined and realistic?
 - Are the team's claims supported by testing, examples, or other evidence from the working prototype?
4. Clarity & Communication (20%)
 - Is the final report well-organized and easy to understand?
 - Is the Demo Day setup engaging and accessible to a business audience?

The final project is graded holistically by the instructor and invited judges based on the final report, the working prototype, and the Demo Day presentation.

Peer Evaluation

A peer evaluation will be conducted at the end of the course for multi-person teams. Each team member rates every other member on a 0–10 scale, and final project grades may be adjusted based on peer evaluation scores:

Score	Meaning
10	Fully contributed: engaged, responsive, constructive, and collaborative.
5	Participated but did not contribute as much as other members.
0	Failed to contribute: missed meetings, unresponsive, or divisive.

Peer scores are averaged and applied as a factor to the final project grade.

Grading

Component	Weight
MiniAssignments (3 x 5%)	15%
Class participation	15%
Major Assignment 1: LLM API & batch text analysis	10%
Major Assignment 2: RAG pipeline	10%
Major Assignment 3: Fine-tuning	10%
Final project	40%
Total	100%

Course Policies

Generative AI Policy

Students are permitted and encouraged to use generative AI tools (e.g., ChatGPT, Claude, Gemini, GitHub Copilot) throughout this course, including assignments, projects, and in-class exercises.

- You are responsible for the accuracy and quality of all submitted work, regardless of whether AI assisted in producing it.
- Submitting AI-generated work that you cannot explain violates academic integrity.

The goal of this course is to develop your ability to work with AI effectively, not to avoid it.

Academic Integrity

I employ the principles and procedures espoused by the University of Washington Student Conduct Code to maintain academic integrity in the course. The Code establishes the expectation that students will practice high standards of professional honesty and integrity. In particular, implementation of the Code at the Foster School of Business prohibits cheating, attempted cheating, and plagiarism, including improper citations of source material, as it pertains to academic work. Suspected violations will be handled in compliance with the University of Washington Student Conduct Code.

Disability-Related Accommodations

To request academic accommodations due to disability, please contact [Disability Resources for Students \(DRS\)](#). If you have a letter from DRS indicating that you have a disability that requires

academic accommodations, please present the letter to me as soon as possible so that we can discuss the support I can offer you in this class.

Religious Accommodation

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at the [Religious Accommodations Policy](#). Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form.